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(57) Abstract:

The invention provides an IoT-enabled Fire Prevention and Safety System for CNG-based vehicles. The system integrates smoke, flame, temperature, and pressure sensors with a microcontroller to detect fire hazards early. On detecting abnormal conditions, it automatically activates a fire suppression mechanism, alerts passengers through audio-visual signals, and sends real-time notifications to emergency contacts and authorities via GSM, GPS, and Bluetooth. Cloud integration allows remote monitoring and data logging. The system operates even during vehicle power failure using a backup battery, offering an intelligent, automated, and scalable solution to enhance passenger safety and prevent fire-related accidents.

FORM 2

THE PATENTS ACT, 1970
(39 of 1970)

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THE PATENTS RULES, 2003

COMPLETE SPECIFICATION
(See Section 10; rule 13)

Title of the Invention

"Fire Prevention and Safety System for CNG Based Vehicles"

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The following specification particularly describes the invention and the manner in which it is performed.

TECHNICAL FIELD

[0001] The present invention relates to automotive safety systems. More specifically, it concerns an IoT-enabled automated fire prevention and safety system for CNG-based vehicles, designed to detect, suppress, and alert occupants and emergency services in the event of a fire.

BACKGROUND ART

[0002] The following discussion of the background of the invention is intended to facilitate an understanding of the present invention. However, it should be appreciated that the discussion is not an acknowledgment or admission that any of the material referred to was published, known, or part of the common general knowledge in any jurisdiction as of the application's priority date. The details provided herein the background if belongs to any publication is taken only as a reference for describing the problems, in general terminologies or principles or both of science and technology in the associated prior art.

[0003] CNG-based vehicles are susceptible to fire accidents due to fuel leakage, electrical short circuits, engine overheating, and collisions. Conventional fire safety measures, such as manual fire extinguishers, are inadequate as they rely on human intervention, which may fail during panic situations. Current solutions lack automated hazard detection, real-time alerts, and suppression mechanisms, leading to increased risk of fatalities, injuries, and property damage. There is a critical need for an intelligent, automated system capable of early fire detection, instant response, and reliable communication with emergency services.In light of the foregoing, there is a need for present invention that overcomes problems prevalent in the prior art associated with the traditionally available method or system, of the above-mentioned inventions that can be used with the presented disclosed technique with or without modification.

[0004] All publications herein are incorporated by reference to the same extent as if each individual publication or patent application were specifically and individually indicated to be incorporated by reference. Where a definition or use of a term in an incorporated reference is inconsistent or contrary to the definition of that term provided herein, the definition of that term provided herein applies, and the definition of that term in the reference does not apply.

OBJECTS OF THE INVENTION

[0005] The primary objectives of the invention are:

- To provide real-time monitoring of CNG-based vehicles using IoT-enabled sensors.
- To enable early detection of fire hazards, including smoke, flame, abnormal temperature, and pressure.
- To automate fire suppression without human intervention.
- To alert passengers immediately through audio-visual warnings.
- To send real-time alerts and location data to emergency services and registered contacts via GSM, GPS, and Bluetooth.

- To provide a cost-effective, scalable, and reliable solution for vehicle fire safety.

[0006] The foregoing and other objects of the present invention will become readily apparent upon further review of the following detailed description of the embodiments as illustrated in the accompanying drawings.

SUMMARY OF THE INVENTION

[0007] The invention provides an automated Fire Prevention and Safety System for CNG-based vehicles that integrate IoT sensors (smoke, flame, temperature, and pressure), a microcontroller-based processing unit, fire suppression mechanisms, and communication modules. On detecting abnormal conditions, the system triggers instant audio-visual alerts, activates a miniature fire suppression unit, and sends real-time notifications to emergency contacts and authorities. Cloud integration allows logging and monitoring for further analysis. The system operates even during vehicle power failure via a backup battery and is suitable for large-scale deployment in vehicles, ensuring enhanced passenger safety and rapid response during fire incidents.

[0008] While the invention has been described and shown with reference to the preferred embodiment, it will be apparent that variations might be possible that would fall within the scope of the present invention.

BRIEF DESCRIPTION OF DRAWINGS

[0009] So that the manner in which the above-recited features of the present invention can be understood in detail, a more particular description of the invention, briefly summarized above, may have been referred by embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this invention and are therefore not to be considered limiting of its scope, for the invention may admit to other equally effective embodiments.

[0010] These and other features, benefits, and advantages of the present invention will become apparent by reference to the following text figure, with like reference numbers referring to like structures across the views, wherein:

[0011] No Figure.

DETAILED DESCRIPTION OF THE INVENTION

[0012] While the present invention is described herein by way of example using embodiments and illustrative drawings, those skilled in the art will recognize that the invention is not limited to the embodiments of drawing or drawings described and are not intended to represent the scale of the various components. Further, some components that may form a part of the invention may not be illustrated in certain figures, for ease of illustration, and such omissions do not limit the embodiments outlined in any way. It should be understood that the drawings and the detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the invention is to cover all modifications, equivalents, and alternatives falling within the scope of the present invention as defined by the appended claim.

[0013] As used throughout this description, the word "may" is used in a permissive sense (i.e. meaning having the potential to), rather than the mandatory sense, (i.e. meaning must). Further, the words "a" or "an" mean "at least one” and the word “plurality” means “one or more” unless otherwise mentioned. Furthermore, the terminology and phraseology used herein are solely used for descriptive purposes and should not be construed as limiting in scope. Language such as "including," "comprising," "having," "containing," or "involving," and variations thereof, is intended to be broad and encompass the subject matter listed thereafter, equivalents, and additional subject matter not recited, and is not intended to exclude other additives, components, integers, or steps. Likewise, the term "comprising" is considered synonymous with the terms "including" or "containing" for applicable legal purposes. Any discussion of documents, acts, materials, devices, articles, and the like are included in the specification solely for the purpose of providing a context for the present invention. It is not suggested or represented that any or all these matters form part of the prior art base or were common general knowledge in the field relevant to the present invention.

[0014] In this disclosure, whenever a composition or an element or a group of elements is preceded with the transitional phrase “comprising”, it is understood that we also contemplate the same composition, element, or group of elements with transitional phrases “consisting of”, “consisting”, “selected from the group of consisting of, “including”, or “is” preceding the recitation of the composition, element or group of elements and vice versa.

[0015] The present invention is described hereinafter by various embodiments with reference to the accompanying drawing, wherein reference numerals used in the accompanying drawing correspond to the like elements throughout the description. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiment set forth herein. Rather, the embodiment is provided so that this disclosure will be thorough and complete and will fully convey the scope of the invention to those skilled in the art. In the following detailed description, numeric values and ranges are provided for various aspects of the implementations described. These values and ranges are to be treated as examples only and are not intended to limit the scope of the claims. In addition, several materials are identified as suitable for various facets of the implementations. These materials are to be treated as exemplary and are not intended to limit the scope of the invention.

[0016] The invention is an IoT-based smart vehicle fire detection and suppression system designed to provide real-time hazard monitoring, automated response, and remote communication to enhance passenger safety. The system integrates data acquisition and sensors, including smoke, flame, temperature, and pressure sensors, strategically placed in critical areas of the vehicle to continuously monitor environmental conditions. Sensor data is transmitted to a processing unit, such as an Arduino UNO or Raspberry Pi, which analyzes the readings against predefined thresholds to detect potential fire events, including abnormal smoke, flames, temperatures exceeding 60°C, or pressure anomalies. Upon hazard detection, the system triggers an automated fire suppression mechanism, such as a miniature CO_2 extinguisher, while simultaneously alerting passengers through buzzers, LED indicators, and display panels. A communication module ensures real-time notifications: GSM transmits SMS alerts to emergency contacts, GPS shares the vehicle's precise location, and Bluetooth provides local alerts in low-network areas like tunnels or underground parking. All events are logged in a database, with cloud integration via IoT platforms like ThingSpeak or Blynk enabling remote monitoring, historical analysis, and predictive improvements. The system is designed for high reliability, with a backup battery ensuring continuous operation during vehicle power failure and calibrated sensors for accurate detection. By addressing the limitations of traditional vehicle fire safety—such as dependence on human intervention, delayed detection, absence of automated suppression, and lack of communication with emergency services—the invention provides early hazard detection, automatic suppression, continuous operation, and scalable, cost-effective deployment suitable for various vehicle types beyond CNG-based systems.

[0017] Various modifications to these embodiments are apparent to those skilled in the art from the description and the accompanying drawings. The principles associated with the various embodiments described herein may be applied to other embodiments. Therefore, the description is not intended to be limited to the 5 embodiments shown along with the accompanying drawings but is to be providing the broadest scope consistent with the principles and the novel and inventive features disclosed or suggested herein. Accordingly, the invention is anticipated to hold on to all other such alternatives, modifications, and variations that fall within the scope of the present invention and appended claims.

CLAIMS

We Claim:

1) A Fire Prevention and Safety System for CNG-based vehicles comprising:

- IoT-enabled sensors for smoke, flame, temperature, and pressure monitoring;
- A microcontroller configured to receive sensor inputs and detect hazardous conditions;
- An automated fire suppression mechanism activated upon detection of fire;
- Audio-visual alert system for passengers;
- Communication modules including GSM, GPS, and Bluetooth for real-time alerts and location sharing.

2) The system of claim 1, wherein the microcontroller compares sensor data against pre-defined thresholds to identify potential fire hazards.

3) The system of claim 1, wherein the fire suppression mechanism comprises a miniature CO_2 extinguisher or automated release system.

4) The system of claim 1, wherein the communication modules transmit alerts to emergency services and registered contacts in real time.

5) The system of claim 1, further comprising cloud integration for data logging, remote monitoring, and analysis.

Dated this27 th day of November, 2025

Pooja

IN/PA/1838

Agent for the Applicant

ABSTRACT

“Fire Prevention and Safety System for CNG Based Vehicles”

The invention provides an IoT-enabled Fire Prevention and Safety System for CNG-based vehicles. The system integrates smoke, flame, temperature, and pressure sensors with a microcontroller to detect fire hazards early. On detecting abnormal conditions, it automatically activates a fire suppression mechanism, alerts passengers through audio-visual signals, and sends real-time notifications to emergency contacts and authorities via GSM, GPS, and Bluetooth. Cloud integration allows remote monitoring and data logging. The system operates even during vehicle power failure using a backup battery, offering an intelligent, automated, and scalable solution to enhance passenger safety and prevent fire-related accidents.

Dated this 27th day of November, 2025

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